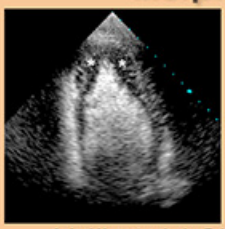
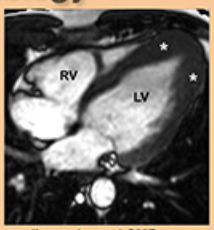
ASE American Society of Echocardiography
Heart & Circulation Ultrasound Specialists

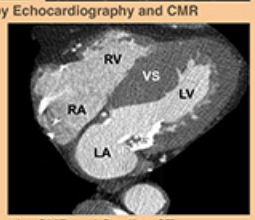
Multimodality Cardiovascular Imaging of Patients with Hypertrophic Cardiomyopathy

Morphology

Apical Hypertrophy by Echocardiography and CMR





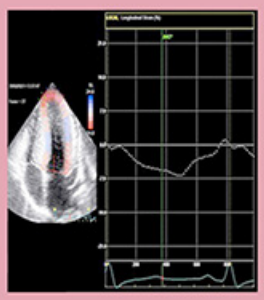
Septal Hypertrophy by CMR and Cardiac CT

- Echocardiography is the initial imaging modality.
- CMR should be used with patients with suboptimal echocardiographic images, and in situations with incomplete and/or unsatisfactory assessment of individual segmental wall thickness by echocardiography. CMR may be considered in selected patients with high index of suspicion for HCM.
- Cardiac CT is recommended when echocardiographic images are inadequate, and when CMR is contraindicated as in patients with ICD/pacemakers.
- Echocardiographic reports should include measurements of LV dimensions, wall thickness (including maximum wall thickness), pattern of hypertrophy and its severity and distribution.

★: site of apical hypertrophy
Top left image acquired after intravenous contrast injection.

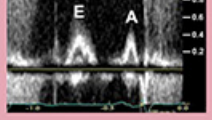
Systolic & Diastolic Function

- Echocardiography is the initial imaging modality of choice for evaluation of LV EF, which should be included in the report.
- CMR is recommended when sub-optimal echocardiographic images are present.
- Cardiac CT or radionuclide angiography can be considered for EF assessment when echocardiographic images are inadequate and CMR is contraindicated.
- Echocardiography is the only modality recommended for evaluation of LV diastolic function and a comprehensive approach should be followed per the recent ASE/EAE diastolic function guidelines.

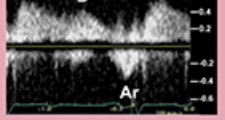


LV global longitudinal strain by speckle tracking echocardiography (STE), should be used in patients with HCM and hyperdynamic LV. LV global strain is markedly reduced at 7%. AVC (aortic valve closure).

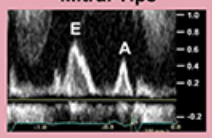
Mitral Annulus



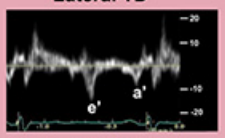
Pulmonary Veins



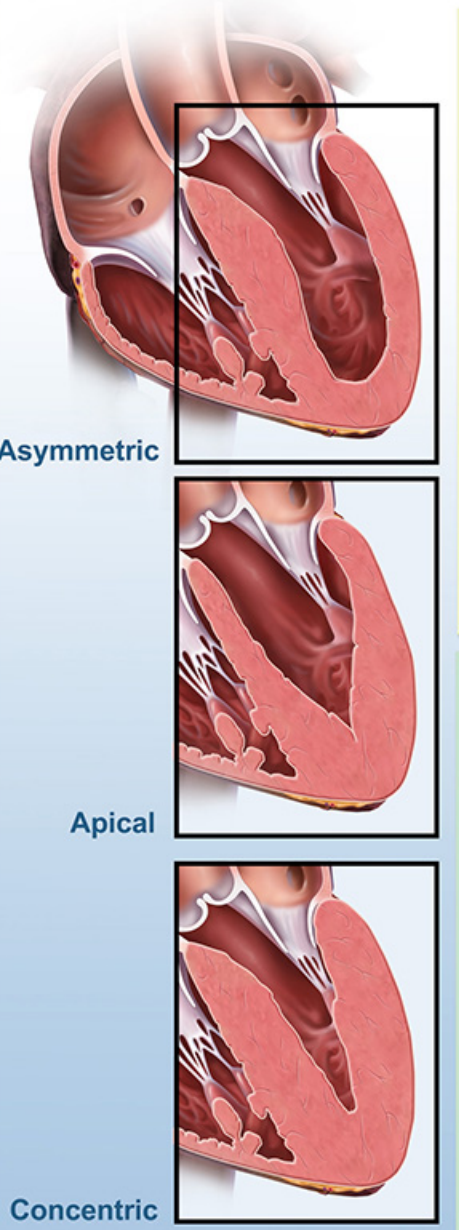
Mitral Tips



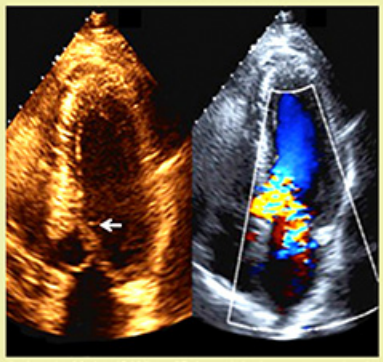
Lateral TD



Assessment of LV diastolic function in a patient with HCM who has an elevated LV end diastolic pressure (EDP), but a normal left atrial (LA) pressure. Mitral inflow shows a short mitral A duration at the level of the mitral annulus, whereas the Ar velocity in pulmonary venous flow is increased in amplitude and duration. Lateral annulus e' velocity is normal and the ratio of peak E velocity (at the level of mitral tips) to e' velocity is <8, consistent with a normal LA pressure. E: peak mitral early diastolic velocity, A: peak mitral late diastolic velocity, S: systolic velocity in pulmonary veins, D: diastolic velocity in pulmonary veins, and Ar: atrial reversal signal in pulmonary veins, e': early diastolic TD velocity, a': late diastolic TD velocity



LVOT Obstruction



Systolic Anterior Motion (SAM) on 2D (arrow). In the same panel, color Doppler shows the high velocities across the LV outflow tract in mosaic color and the eccentric mitral regurgitation jet that is directed posterolaterally.

- Echocardiography is the recommended test. PW Doppler is used to localize site of obstruction and CW Doppler is needed to determine peak gradient.
- In patients with LVOT gradient <30 mmHg at rest, gradients can be provoked by Valsalva, amyl nitrite (when available), and if the patient is symptomatic with treadmill exercise.
- CMR may be considered in more challenging clinical scenarios as in patients with suspected subvalvular pathology or those with previous intervention.

CAD Diagnosis

- In HCM patients with chest pain and low probability of CAD, stress SPECT imaging can be considered.
- Coronary angiography, including CT angiography is recommended in patients with chest pain and intermediate or high pre-test probability of CAD.

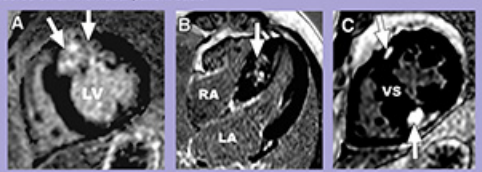


SPECT perfusion imaging from an HCM patient. Septal thickness is increased as is the count activity (hot spot) in the septum relative to the lateral wall. The computer analysis software registered a spurious fixed perfusion defect (scar) in the lateral and apical regions upon normalization to the septum (SA: short axis, HLA: horizontal long axis, VLA: vertical long axis).

Imaging Risk Factors of SCD

As part of a comprehensive SCD risk assessment that also includes medical history and family/genetic history, the following imaging specific variables may be considered on an individual patient basis.

Risk Factor	Imaging Modality
Maximum wall thickness ≥3 cm	Echocardiography, CMR, cardiac CT
End stage HCM: EF <50%	Echocardiography, radionuclide angiography, CMR, cardiac CT
Apical aneurysms	Contrast echocardiography, CMR, and cardiac CT
LVOT gradient ≥ 30mmHg	Doppler echocardiography
Perfusion defects	SPECT (though no association in some studies)
Reduced coronary flow reserve	PET (observations limited to very few patients)
LGE (presence and extent)	CMR (evidence not conclusive)



Contrast-enhanced CMR with late gadolinium enhancement (arrows) in HCM.

A. Asymptomatic 58-year-old woman with a large transmural area of late gadolinium enhancement in the basal anterior septum and anterior wall;

B. Diffuse and patchy area of late gadolinium enhancement in the mid-myocardial area of ventricular septum in a 21-year-old man;

C. Late gadolinium enhancement confined to the area of RV free wall insertion into the anterior and posterior ventricular septum.

Adapted from: Nagueh SF, Bierig SM, Budoff MJ, Desai M, Dilsizian V, Eidem B, Goldstein SA, Hung J, Maron MS, Ommen SR, Woo A, American Society of Echocardiography Clinical Recommendations for Multimodality Cardiovascular Imaging of Patients with Hypertrophic Cardiomyopathy, J Am Soc Echocardiogr 2011; 24:473-98.
Endorsed by the American Society of Nuclear Cardiology, Society for Cardiovascular Magnetic Resonance, and Society of Cardiovascular Computed Tomography.

Poster ordering information and full text of ASE guideline documents available at: www.asecho.org/guidelines

The information on this poster does not constitute the offering of medical advice by ASE, and should not be used as the sole basis for medical practice decisions.
Design and illustration by **medmovie.com** © Copyright 2011 The American Society of Echocardiography